Lecture 10: Black-Scholes Equation

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Based on lecture notes by me and many previous CS335 instructors

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Announcements:

1. There will be no In-Class Activity on Wednesday October 21 and Monday October 26.
2. There will be no class on the day of the midterm, Wednesday October 28. Instead I'll run an office hour. Please drop by to my office, MC 6431, if you have any questions.
3. Office hours for both A2 and the midterm will be announced shortly on Piazza.
4. Check the announcements I have posted on both Piazza and D2L.
5. Let me know if you have any questions.

Deriving the *Black-Scholes equation*

Black-Scholes is a (deterministic) PDE governing the *exact* price of European options, under a certain set of assumptions:

- The Stock price follows geometric Brownian motion (GBM)
- Risk-free rate of return is known
- A hedger (e.g., us) can borrow/lend at the risk-free rate r
- There are no transaction costs
- The volatility is known (though not necessarily constant)
- Short selling is allowed, i.e., we can hold negative quantities of assets.

Deriving Black-Scholes

Let's try to derive it by constructing a risk-free portfolio, using our no-arbitrage principle.

As usual, let Π = hedging portfolio, V = option value, S = asset price, and δ = the # of shares of S in Π. Remember, δ is a quantity we can choose/control.

Our portfolio value is $\Pi = V - \delta S$. Consider how the portfolio Π changes over a small time period, $t \to t + dt$, i.e., $\Pi \to \Pi + d\Pi$.

$$d\Pi = dV - \delta dS$$

We form the portfolio at time t, with δ shares. S evolves randomly, but δ is held constant over $[t, t + dt]$. (That's why there's no $d\delta$ term above.)

Deriving Black-Scholes

We want to work out an expression for $d\Pi = dV - \delta dS$.

We assume S follows GBM as before, so:

$$dS = a\,dt + b\,dZ$$

with $dZ = \phi\sqrt{dt}$, $\phi \sim N(0,1)$, $a = \mu S, b = \sigma S$.

What about the option price V?

V depends on stock price S, and we know dS, so we can use Ito's Lemma!

Deriving Black-Scholes

$V = V(S,t)$ so we directly apply Ito's Lemma to find:

$$dV = \left(aV_S + \frac{b^2}{2}V_{SS} + V_t\right)dt + bV_S\,dZ$$

$$= \left(\mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} + V_t\right)dt + (\sigma S V_S)dZ$$

Now, plug dS and dV back into $d\Pi = dV - \delta dS$:

$$d\Pi = dV - \delta\mu S\,dt - \delta\sigma S\,dZ$$

$$= \left(\mu S V_S - \delta\mu S + \frac{1}{2}\sigma^2 S^2 V_{SS} + V_t\right)dt + (\sigma S V_S - \delta\sigma S)\,dZ$$

Deriving Black-Scholes

$V_S = \delta \Rightarrow \mu S V_S - \mu S V_S = 0$

This is now an SDE for portfolio value.

$$d\Pi = \left(\mu S V_S - \delta \mu S + \frac{1}{2}\sigma^2 S^2 V_{SS} + V_t\right) dt + (\sigma S V_S - \delta \sigma S)\, dZ$$

We want the portfolio evolution in a risk-free way (to exploit no-arbitrage). How?

By ensuring the random term, $(\sigma S V_S - \delta \sigma S) dZ$, is zero, we completely eliminate randomness! i.e., set $\delta = V_S$

We're left with $d\Pi = \left(\frac{1}{2}\sigma^2 S^2 V_{SS} + V_t\right) dt$

Black-Scholes equation

Since Π is risk-free now, no-arbitrage principle says that it must earn the risk-free rate r. i.e., change in value is just interest rate r times the current value Π, times the amount of time dt:

$$d\Pi = r\Pi\,dt = r(V - \delta S)dt = r(V - V_S S)dt$$

But we also saw (via Ito's lemma) that:

$$d\Pi = \left(\frac{1}{2}\sigma^2 S^2 V_{SS} + V_t\right)dt$$

Equating these two expressions for $d\Pi$, we arrive at the Black-Scholes PDE!

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$$

Black-Scholes equation

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$$

This tells us how the option value V evolves over time (V_t), based on the option value V, V's derivatives w.r.t S, the risk-free rate r, stock volatility σ, and the stock value S.

Note: This equation is independent of the drift rate, μ, of the stock! (Similar to how our no-arbitrage lattice didn't depend on true up/down probabilities).

First published in 1973 - Scholes & Merton received the Nobel Prize for it in 1997. Sadly, Black had already died in 1995.

Black-Scholes

This (continuous) equation is solved *backwards* in time from the payoff at $t = T$ back to $t = 0$.

In this aspect, it's similar to the (discrete) lattice pricing approached & dynamic programming (e.g., marriage problem).

Notice: We set $\delta = V_S = \frac{\delta V}{\delta S}$ under Black-Scholes. When we derived our lattice pricing approach, we had used $\delta_j^n = \frac{V_{j+1}^{n+1} - V_j^{n+1}}{S_{j+1}^{n+1} - S_j^{n+1}}$

As $\Delta t \to 0$, these δ expressions agree.

Hedging in *Continuous* Time

Suppose we short (borrow and sell) an option worth V at $t = 0$ (which gives us V in cash).

Then we borrow $SV_S - V$ from the back (short, negative) so

$$\text{cash} = (SV_S - V) + V = SV_S$$

Lastly, use this cash to buy V_S shares of the asset (stock) at price S.

Total portfolio is:

Long shares of stock

cash in the bank

$$(-V) + (SV_S) + (V - SV_S) = 0$$

Short position (we sold the option)

Continuous-Time Hedging

$$(-V) + (SV_S) + (V - SV_S) = 0$$

So our total position is still 0.

Note:

- If $V - SV_S < 0$ we have borrowed from bank (short cash position).
- If $V - SV_S > 0$ we have deposited cash in bank (long cash position).

Continuous-Time Hedging

After the small period of time dt, we are at $t + dt$, and $S \to S + dS$, and

$V_S \to \frac{\partial V}{\partial S}(S+dS, t+dt)$

We then rebalance the hedge. i.e., we *instantaneously* buy/sell shares so that we own $\frac{\partial V}{\partial S}(S+dS, t+dt)$ shares at $t + dt$.

i.e., update the number of shares to match the updated value of $\frac{\partial V}{\partial S}$.

We borrow/deposit cash in the bank accordingly to compensate.

Continuous-Time Hedging

This is called a dynamic hedge.

Cash continuously flows in/out of the bank in response to changes in the value of S.

At any instant in time, we can always liquidate our position (portfolio) payoff any obligations from the short position, pay back bank loans, and come out to zero gain/loss, regardless of S's value.

This strategy is self-financing; give the initial cash from the sale (shorting) of the option, no further cash is required.

Option Price Notes

- The option's price is dynamic; it changes in response to changing S.
- The option price determined by Black-Scholes is the option's market price at any given time.
- Black-Scholes is *not* the expected payoff value of the option.
- The preceding discussion applies only to European option.

End of Lecture 10

Midterm Cut Off